



UNIVERSITY OF THE PUNJAB

Fourth Semester - 2018

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Statistics-IV

Course Code: STAT-203 / STT-22314 Part - II

TIME ALLOWED: 2 Hrs. & 45 Mins.

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q2. Differentiate between the following terms:

(4 marks each)

- Regression and Correlation
- One-way classification and Two-way Classification
- Completely Randomized Design and Completely Randomized Block Design
- Mean Prediction Interval Estimates and Individual Prediction Interval Estimates
- Multiple Correlation Coefficient and Partial Correlation Coefficient.

SUBJECTIVE

Q3. Three sections of the same statistics course are taught by three teachers. The final grades given by the three teachers (independently) were recorded as follows: (06)

Teacher A	75	91	84	45	82
Teacher B	59	83	99	77	65
Teacher C	66	77	51	90	73

Test the hypothesis that the average grades given by the three teachers are same.

Q4. Is the type of beverage ordered with lunch at a restaurant independent of the age of the customer? A random poll of 55 lunch customers is taken, resulting in the following contingency table of observed values. Use $\alpha = .01$ to determine whether the two variables are independent. (06)

		Preferred Beverage		
		Coffee/Tea	Soft Drink	Other (Milk etc.)
Age	1 - 34	8	14	9
	35 - 55	5	4	2
	> 55	6	4	3

Q5. Five varieties of Rice are compared in five randomized complete blocks. The SS for Total, Blocks and Varieties are 282.30, 26.96, and 155.44 respectively. (06)

- Test the significance of difference between varietal means
- Compare the efficiency of this design with the design in which blocks are ignored.

Q6. Given that:

(12)

X_1	3	5	6	8	12	14
X_2	16	10	7	4	3	2
X_3	90	72	54	42	30	12

- Determine r_{12} , $r_{12.34}$ and $R_{1.23}$
- Test each one of these correlation coefficients for a significance at 5% level.

question 2(1)

Q7.2 Write short answers to the following questions in the space provided underneath.

Q.1. Differentiate between regression analysis and correlation analysis.

(Federal Board Islamabad 2010)

Ans. In the simple regression analysis we obtain a regression equation which can be used to estimate the values of the dependent variable on the basis of the independent variable whose values are known. In correlation analysis we study the degree of closeness of relationship between the variables. In case of regression the dependent variable is assumed to be *fixed* or *known*; in case of correlation both the variables are assumed to be *random*. Correlation simply means the co-variation of variables. It does not mean that changes in one variable cause a change in the other.

way classification is defined. The first two classification procedures for the first two classifications only as a one-way analysis of variance and a two-way analysis of variance respectively.

20.2 ONE-WAY ANALYSIS OF VARIANCE

The *one-way analysis of variance* is also called the *one-variable-of-classification analysis of variance*. The data are classified into k classes or groups, etc. on the basis of a single criterion. The technical term for a class, a group or a variable of classification is *treatment*.

will impair the technique. However, investigations have shown that minor deviations from normality and equality of variances are to be tolerated. Sometimes, an appropriate transformation such as using the square root or logs, etc., is made to satisfy these assumptions.

20.3 TWO-WAY ANALYSIS OF VARIANCE

When each observation is classified according to two criteria (or variables) of classification simultaneously, we use the two-way analysis of variance technique. The classified data are recorded in a table, in which the columns represent one criterion (or variable) of the classification and the rows represent the other criterion. If there are c -columns and r -rows in the table, then there will be altogether rc cells. Each cell may contain a single observation or several observations.

There are two basic forms of two-way analysis of variance, depending upon whether the two variables of classification are independent or whether they interact. Two variables (or criteria) of classification are said to *interact* when they *together* have an added effect that they do not have individually. For example, suppose that we classify salesmen according to, say, age and educational attainment with a view to determining whether age and education have significant effect on the

3 Completely Randomized Design & Completely Randomized Block Design? (04)

A complete randomized (CR) designed which is the basic designed, may be defined as a design in which the treatment are assigned to be experimental units completely at random that is the randomization is done without any restriction. The complete randomized designed is most useful in situation (1) The experimental units are homogenous (2) The experiments are small such as laboratory experiment.

Completely Randomized block designed:-

- The Complete Randomized block designed are
- 1) The experimental material is divided into group & ~~at~~ ^{into} blocks such a manner that the experimental unit with in particular block are relatively homogenous.
 - 2) Each block contain a complete set of observation.
 - 3) The treatment are assigned at random to experimental unit with in each block, which mean the randomization is restricted with in blocks.

Q Describe Bartlett's test of homogeneity of variance?

The Bartlett's test of homogeneity is

$$U = 2.30269$$

C_{10}

11.3 MULTIPLE CORRELATION CO-EFFICIENT

The *co-efficient of multiple correlation* measures the degree of relationship between a variable and its estimate from the regression equation. In other words, it is a product moment correlation between a variable, say x_1 , and its value estimated by the regression equation $x_1 = b_{12.3} x_2 + b_{13.2} x_3$. The co-efficient of multiple correlation between x_1 and the variables x_2 and x_3 combined, is denoted symbolically by $R_{1.23}$.

Let us denote the estimated value of x_1 by $\hat{x}_1$. Then by definition,

$$R_{1.23} = \frac{\text{Cov}(x_1, \hat{x}_1)}{\sqrt{\text{Var}(x_1) \text{Var}(\hat{x}_1)}} = \frac{\sum x_1 \hat{x}_1}{\sqrt{\sum x_1^2 \sum (\hat{x}_1)^2}} \quad \text{question 2(5)}$$

$$\begin{aligned} \text{Now } \sum x_1 \hat{x}_1 &= \sum x_1 (x_1 - x_{1.23}) & (\because \hat{x}_1 = x_1 - x_{1.23}) \\ &= \sum x_1^2 - \sum x_1 x_{1.23} \\ &= \sum x_1^2 - \sum x_{1.23} x_{1.23} & (\because \sum x_1 x_{1.23} = \sum x_{1.23} x_{1.23}) \\ &= n(S_1^2 - S_{1.23}^2), \end{aligned}$$

$$\sqrt{0.64} = \sqrt{0.5727} = 0.757$$

11.4 PARTIAL CORRELATION

A *partial correlation* measures the degree of linear relationship between any two variables in a multivariable problem under the condition that any common relationship or influence with all other variables (or some of them) has been removed. Stated differently, if there are three variables X_1 , X_2 and X_3 , then the correlation between X_1 and X_2 after removing the linear effect of X_3 from X_1 and from X_2 , is the *partial correlation*. The sample co-efficient of partial correlation measuring the strength of the relationship (correlation) between X_1 and X_2 , when the influence of X_3 has been removed, is denoted symbolically by $r_{12.3}$. By removing the influence, we mean subtracting the fitted regression $\hat{X}_1$ from the observed values X_1 , obtaining the residual – a part of X_1 not explained by X_3 .

To derive the co-efficient of partial correlation $r_{12.3}$, we use the variables x_1 , x_2 and x_3 which are deviations from their means. The linear regression of x_1 on x_3 and of x_2 on x_3 are $x_1 = b_{13}x_3$ and $x_2 = b_{23}x_3$. Removing the linear effect of x_3 from x_1 and from x_2 and denoting the residuals by $x_{1.3}$ and $x_{2.3}$, we get

$$x_{1.3} = x_1 - b_{13}x_3, \text{ and } x_{2.3} = x_2 - b_{23}x_3.$$

These residuals may be written as

$$x_{1.3} = x_1 - r_{13} \frac{S_1}{S_3} x_3, \text{ and } x_{2.3} = x_2 - r_{23} \frac{S_2}{S_3} x_3.$$

Now the co-efficient of partial correlation is the product moment correlation co-efficient between residuals $x_{1.3}$ and $x_{2.3}$. Thus by definition

$$r_{12.3} = \frac{\sum x_{1.3} x_{2.3}}{\sqrt{\sum x_{1.3}^2 \sum x_{2.3}^2}}$$

question 2(5)

Multiple correlation

- It is the simple correlation b/w variable and as estimate of that variable obtained from a linear equation involving two or more independent variables.

- Denoted by R .

Partial correlation

- It is the simple correlation b/w variable and one independent variable after adjusting each for the effect of one or more other variables.

- Denoted by r_{xy} .

question 2(5)

to measure coefficient of mean-square contingency

Three sections of the same statistic course are taught

Q by three teachers The final grades given by the three teachers
Test the hypothesis (independence) were recorded as follows
(2018)

① Hypotheses-

H_0 = All means are equal

H_1 = Not all means equal

② $1 - \alpha = 0.05$

③ Test statistic - $F = \frac{S_1^2}{S_2^2}$

④ Calculations-

	75	91	84	45	82
	59	83	99	77	65
	66	77	51	90	73
ΣT_j	200	251	234	212	220
ΣT_j^2	40000	63001	54756	44944	48400
$\Sigma \Sigma x_{ij}^2$	13462	21099	19458	16054	16278

$$\text{Correlation coefficient} = \frac{(\Sigma T_j)^2}{n} = \frac{(1797)^2}{15} = 83179.26$$

$$\text{B/w S.S} = \text{Total S.S} = \Sigma \Sigma x_{ij}^2 - C.F = \frac{86351}{3} - 83179.26 = 3171.73$$

$$\text{B/w S.S} = \frac{\Sigma T_j^2}{3} - C.F = \frac{251101}{3} - 83179.26 = 521.073$$

$$\text{Wthins S.S} = \text{Tot w S.S} - \text{B/w S.S} = 3171.73 - 521.073 = 2650.65$$

11.5. (a) The estimated multiple regression of X_3 on X_1 and X_2 is

$$\hat{X}_3 = b_{3.12} + b_{31.2}X_1 + b_{32.1}X_2$$

where b 's are the least squares estimates of population parameters.

The three normal equations are

$$\sum X_3 = nb_{3.12} + b_{31.2}\sum X_1 + b_{32.1}\sum X_2,$$

$$\sum X_1X_3 = b_{3.12}\sum X_1 + b_{31.2}\sum X_1^2 + b_{32.1}\sum X_1X_2,$$

$$\sum X_2X_3 = b_{3.12}\sum X_2 + b_{31.2}\sum X_1X_2 + b_{32.1}\sum X_2^2.$$

The calculations needed to find b 's are shown in the table below:

X_1	X_2	X_3	X_1^2	X_2^2	X_3^2	X_1X_2	X_2X_3	X_1X_3
3	16	90	9	256	8100	48	1440	270
5	10	72	25	100	5184	50	720	360
6	7	54	36	49	2916	42	378	324
8	4	42	64	16	1764	32	168	336
12	3	30	144	9	900	36	90	360
14	2	12	196	4	144	28	24	168
48	42	300	474	434	19008	236	2820	1818

Substituting the sums in the normal equations, we get

$$6b_{3.12} + 48b_{31.2} + 42b_{32.1} = 300$$

$$48b_{3.12} + 474b_{31.2} + 236b_{32.1} = 1818$$

$$42b_{3.12} + 236b_{31.2} + 434b_{32.1} = 2820$$

Solving them simultaneously, we obtain

$$b_{3.12} = 61.3993, b_{31.2} = -3.6461 \text{ and } b_{32.1} = 2.5385$$

Hence the desired estimated regression equation of X_3 on X_1 and X_2 is

$$\hat{X}_3 = 61.40 - 3.65X_1 + 2.54X_2.$$

(b) When $X_1 = 10$ and $X_2 = 6$, we get

$$\begin{aligned} X_3 &= 61.40 - 3.65(10) + 2.54(6) \\ &= 61.40 - 36.5 + 15.24 = 40. \end{aligned}$$

(c) The multiple correlation co-efficient is the positive square root of $R^2_{3.12}$, where

$$\begin{aligned} R^2_{3.12} &= \frac{\sum(\hat{X}_3 - \bar{X}_3)^2}{\sum(X_3 - \bar{X}_3)^2} \\ &= \frac{b_{3.12}\sum X_3 + b_{31.2}\sum X_1X_3 + b_{32.1}\sum X_2X_3 - (\sum X_3)^2/n}{\sum X_3^2 - (\sum X_3)^2/n} \end{aligned}$$

question 6

$$\begin{aligned} &= \frac{(61.3993)(300) + (-3.6461)(1818) + (2.5385)(2820) - (300)^2/6}{19008 - (300)^2/6} \\ &= \frac{18419.79 - 6628.6 + 7158.57 - 15000}{19008 - 15000} \\ &= \frac{3949.75}{4008} = 0.9855 \end{aligned}$$

$$\text{Hence } R_{3.12} = \sqrt{0.9855} = 0.99.$$

$$= \frac{3949.75}{4008} = 0.9855$$

Hence $R_{3,12} = \sqrt{0.9855} = 0.99$.

question 6

Alternatively:

$$\text{Now, } \bar{X}_1 = \frac{\sum X_1}{n} = \frac{48}{6} = 8; \bar{X}_2 = \frac{\sum X_2}{n} = \frac{42}{6} = 7;$$

$$\bar{X}_3 = \frac{\sum X_3}{n} = \frac{300}{6} = 50.$$

$$S_1 = \sqrt{\frac{\sum X_1^2}{n} - \left(\frac{\sum X_1}{n}\right)^2} = \sqrt{\frac{474}{6} - \left(\frac{48}{6}\right)^2}$$

$$= \sqrt{79 - 64} = \sqrt{15} = 3.87;$$

$$S_2 = \sqrt{\frac{\sum X_2^2}{n} - \left(\frac{\sum X_2}{n}\right)^2} = \sqrt{\frac{434}{6} - \left(\frac{42}{6}\right)^2}$$

$$= \sqrt{72.33 - 49} = \sqrt{23.33} = 4.83;$$

$$S_3 = \sqrt{\frac{\sum X_3^2}{n} - \left(\frac{\sum X_3}{n}\right)^2} = \sqrt{\frac{19008}{6} - \left(\frac{300}{6}\right)^2}$$

$$= \sqrt{3168 - 2500} = \sqrt{668} = 25.84;$$

and

$$r_{12} = \frac{\sum X_1 X_2 - n \bar{X}_1 \bar{X}_2}{\sqrt{[\sum X_1^2 - n \bar{X}_1^2] [\sum X_2^2 - n \bar{X}_2^2]}}$$

$$= \frac{236 - 6(8)(7)}{\sqrt{[474 - 6(64)] [434 - 6(49)]}} = \frac{236 - 336}{\sqrt{(90)(140)}}$$

$$= \frac{-100}{112.25} = -0.89;$$

$$r_{23} = \frac{\sum X_2 X_3 - n \bar{X}_2 \bar{X}_3}{\sqrt{[\sum X_2^2 - n \bar{X}_2^2] [\sum X_3^2 - n \bar{X}_3^2]}}$$

$$= \frac{2820 - 6(7)(50)}{\sqrt{[434 - 6(49)] [19008 - 6(2500)]}} = \frac{2820 - 2100}{\sqrt{(140)(4008)}}$$

$$= \frac{720}{749.08} = 0.96;$$

$$r_{31} = \frac{\sum X_1 X_3 - n \bar{X}_1 \bar{X}_3}{\sqrt{[\sum X_1^2 - n \bar{X}_1^2] [\sum X_3^2 - n \bar{X}_3^2]}}$$

$$= \frac{1818 - 2400}{\sqrt{(90)(4008)}} = \frac{-582}{600.6} = -0.97.$$

$$r_{31} = \frac{\sum X_1 X_3 - n \bar{X}_1 \bar{X}_3}{\sqrt{[\sum X_1^2 - n \bar{X}_1^2] [\sum X_3^2 - n \bar{X}_3^2]}}$$

$$= \frac{1818 - 2400}{\sqrt{(90)(4008)}} = \frac{-582}{600.6} = -0.97.$$

(a) The linear regression equation of x_3 on x_1 and x_2 is

$$\frac{x_3}{S_3} = \left(\frac{r_{32} - r_{13}r_{12}}{1 - r_{12}^2} \right) \frac{x_2}{S_2} + \left(\frac{r_{31} - r_{23}r_{12}}{1 - r_{12}^2} \right) \frac{x_1}{S_1},$$

where $x_1 = X_1 - \bar{X}_1$, $x_2 = X_2 - \bar{X}_2$ and $x_3 = X_3 - \bar{X}_3$.

Substituting the values, we get

$$\frac{x_3}{25.84} = \left(\frac{(0.96) - (-0.97)(-0.89)}{1 - (-0.89)^2} \right) \frac{x_2}{4.83} + \left(\frac{-0.97 - (0.96)(-0.89)}{1 - (-0.89)^2} \right) \frac{x_1}{3.87}$$

$$\text{Or } \frac{x_3}{25.84} = \left(\frac{0.0967}{0.2079} \right) \frac{x_2}{4.83} + \left(\frac{-0.1156}{0.2079} \right) \frac{x_1}{3.87}$$

$$\text{Or } x_3 = 2.49x_2 - 3.71x_1$$

$$\text{Or } (X_3 - 50) = 2.49(X_2 - 7) - 3.71(X_1 - 8)$$

$$\text{Or } X_3 = 62.25 - 3.71X_1 + 2.49X_2.$$

(b) When $X_1 = 10$, and $X_2 = 6$, we get

$$X_3 = 62.25 - 3.71(10) + 2.49(6) = 40$$

question 6

$$(c) R^2_{3.12} = \frac{r_{31}^2 + r_{32}^2 - 2r_{12}r_{23}r_{31}}{1 - r_{12}^2}$$

$$= \frac{0.9409 + 0.9216 - 2(-0.89)(0.96)(-0.97)}{1 - 0.7921}$$

$$= \frac{1.8625 - 1.6575}{0.2079} = \frac{0.2050}{0.2079} = 0.9860$$

Hence $R_{3.12} = \sqrt{0.9860} = 0.993$.